Class-Conditional Generative Augmentation for Rare-Class Infant Cry Classification

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Abstract

Infant cry classification is a clinically-adjacent task in which the rarest classes (such as pain) are also the most safety-critical. Public corpora are heavily imbalanced; in the donateacry-corpus, *hungry* accounts for 84% of clips and *burping* for 1.8%. A naive log-mel CNN trained on this corpus achieves 84% aggregate accuracy by collapsing onto the majority class, missing every safety-critical case. We study whether the combination of (i) class-conditional denoising diffusion for rare-class data synthesis and (ii) a small ablation over optimizer recipes (class-weighted cross-entropy, balanced sampler, focal loss) can change this. We deduplicate a second public corpus (Donate-A-Cry-Augmented) at the content level to surface a fairness-relevant data-quality issue: many of the Kaggle redistribution's labeled rare-class clips are byte-identical to donateacry clips of *other* classes, and we filter them out. After integration, we report a controlled $4 \times 3 \times 3$-seed augmentation matrix (36 cells). The best cell, *generative* + *weighted_ce*, achieves macro-F1 = 0.258 (vs. 0.194 baseline; +33%) and *burping* recall = 0.667 (vs. 0 across all 18 non-generative cells), while preserving 81.6% accuracy. *belly_pain* recall remains hard at this data scale (11 real training clips). The role of each axis decomposes cleanly: a larger train pool unblocks the diffusion model's class-conditional learning (sample-quality probe rises from 0.51 to 0.93 on *belly_pain*, 0.00 to 0.08 on *burping*); generative augmentation is the only intervention that flips the argmax for *burping*; the balanced sampler trades majority-class accuracy for rare-class recall. The project is released open source with a datasheet, model card, and pre-registered evaluation harness.

1 Introduction

Audio-based infant cry classification is studied as a possible decision-support input for parental apps and neonatal monitoring. The clinical and ethical risk profile of such a system is dominated by its behavior on rare, safety-critical classes: a missed pain cry has a far higher cost than a missed hunger cue. However, public datasets are dominated by a single majority class, and the standard pattern recognition pipeline can achieve high accuracy by trivially predicting that class for every input. We argue that the relevant evaluation question is not aggregate accuracy or even macro-F1, but per-class recall on rare classes, under realistic noise and out-of-distribution conditions. Synthetic data augmentation has been studied in audio classification, but it has rarely been framed as a fairness intervention targeted at safety-critical rare classes. We close this gap with a small, fully reproducible study and an explicit ethics review.

2 Related Works

We position the work at the intersection of three small literatures. First, *infant cry classification on public corpora* commonly uses donateacry, Baby Chillanto, or proprietary clinical corpora; reported accuracy numbers are typically in the 70–90% range and are dominated by a single majority class. Few prior works disaggregate per-class metrics, so the rare-class failure mode we describe in Section 5 is largely hidden in the literature. Second, *synthetic data augmentation for audio* ranges from SpecAugment Park et al. [2019] (time/frequency masking, well-validated) to vocoder-based augmentation and audio diffusion models. We use a small class-conditional DDPM Ho et al. [2020] with classifier-free guidance Ho and Salimans [2021] and DDIM sampling Song et al. [2021], sized for the donateacry data scale. Third, *class-imbalance interventions in clinically-adjacent ML* include class-weighted losses, balanced sampling, focal loss Lin et al. [2017], and re-balanced fine-tuning. The fairness audit framing we adopt is closest to model cards Mitchell et al. [2019] and datasheets Gebru et al. [2021], with concrete analogous cases drawn from healthcare ML Obermeyer et al. [2019], Sjoding et al. [2020].

3 Data

Primary corpus. The donateacry-corpus Veres [2018] contains 457 caregiver-uploaded clips labeled with five causes: *belly_pain, burping, discomfort, hungry, tired*. We resample to 16 kHz mono, fix duration to 5 s by center-crop or zero-pad, and compute log-mel spectrograms with 64 mel bands, $n_{\text{fft}} = 1024$, hop length 512, producing $(1, 64, 128)$ tensors. Splits are class-stratified at the file level (seed=42, 70/15/15), and a pytest assertion verifies that no test file ever appears in any synthesis or augmentation manifest. The corpus is heavily imbalanced (Table 1); the rarest classes (*burping* with 8 clips, *belly_pain* with 16 clips) are the safety-critical targets.

Second corpus and a data-quality finding. We integrated the Donate-A-Cry-Augmented corpus from Kaggle, which redistributes donateacry under augmented filenames. On md5 content comparison, we found that 73 of 118 "burping" clips, 66 of 127 "belly_pain" clips, 103 of 136 "tired" clips, and 103 of 138 "discomfort" clips are byte-identical to donateacry clips of *other* classes (mostly *hungry*). A model trained naively on the Kaggle CSV labels would learn a correlation between the burping label and what is acoustically a hungry cry. Our manifest builder filters these duplicates out at build time, retaining only clips whose md5 does not appear anywhere in donateacry. After integration, the train pool grows from 320 to 405 clips and rare-class train counts grow from $\{$belly_pain: 11, burping: 6$\}$ to $\{52, 43\}$. Validation and test splits remain donateacry-real-only.

Table 1: donateacry-corpus class counts after stratified split (seed=42), and train-pool counts after kaggle integration.

Class	All	Train (donateacry)	Train (after kaggle)	Val	Test	%
hungry	382	267	267	57	58	83.6
discomfort	27	19	22	4	4	5.9
tired	24	17	21	4	3	5.3
belly_pain	16	11	52	2	3	3.5
burping	8	6	43	1	1	1.8

4 Methods

Baseline classifier. A four-block convolutional network on log-mel input with batch normalization, dropout, and a global-average-pooling head ($\sim$1.2M parameters). Trained with cosine learning rate decay over 30 epochs.

Optimizer recipes. Three orthogonal recipes are evaluated:

- **weighted_ce**: class-weighted cross-entropy with inverse-square-root frequencies, no balanced sampler.

- **balanced**: plain cross-entropy with a weighted-random sampler over the train pool, equalizing per-batch class frequency.
- **focal**: focal loss Lin et al. [2017] with $\gamma = 2$ and the same per-class α as weighted_ce.

Per cell, the only difference between recipes is the loss/sampler combination; everything else (architecture, learning rate, schedule, epochs, augmentations) is held fixed.

Classical augmentation. SpecAugment-style time and frequency masking, time shift, and additive Gaussian noise applied to the standardized log-mel spectrogram with probability 0.8.

Class-conditional diffusion. A small UNet ($\sim$3.8M parameters) with FiLM-conditioned residual blocks and a single self-attention block at the bottleneck resolution. The conditioning vector concatenates a sinusoidal time embedding with a class label embedding (one extra slot reserved as a null class for classifier-free guidance Ho and Salimans [2021]). The DDPM uses 1000 timesteps with a linear β schedule and is trained on the post-integration train split (405 clips). Sampling uses 25-step DDIM Song et al. [2021] with classifier-free guidance scale 2.0. The diffusion model is trained for 30 epochs on a single Apple-MPS device; longer training was attempted but diverged after epoch 31, consistent with the small dataset size.

Augmentation arms.

- **none**: train on the merged train split only.
- **classical**: as above + SpecAugment.
- **generative**: train split + DDPM-sampled spectrograms targeted at *belly_pain* and *burping* (synth-to-real ratio $3\times$).
- **classical+generative**: stack of the above.

4.1 Innovation

We treat generative augmentation as a *fairness intervention*, not a regularizer. Synthetic samples are added only to the rarest classes; the headline metric is rare-class recall and per-class FNR (not macro-F1); and we publish a per-class accountability table alongside aggregate metrics. The class-consistency probe in Section 5 also enables a cheap diagnostic of whether the generative samples carry usable class signal at all, before any downstream classifier sees them.

5 Experiments and Results

5.1 Sample-quality probe

Before training the augmented classifier, we run a class-consistency probe: the naive baseline classifier (trained on real data only, with weighted CE) is run on the synthetic samples and asked to predict their intended class. If the synthetic samples carry a class signal that is at all distinct from *hungry*, the probe should at least sometimes break character.

Training the DDPM on the smaller, donateacry-only train pool (11 belly_pain, 6 burping) gave a probe recall of 0.51 on synthetic *belly_pain* and 0.00 on synthetic *burping*. Re-training on the merged train pool (52 belly_pain, 43 burping) raises probe recall to **0.93** on *belly_pain* and **0.08** on *burping*: the larger rare-class signal is the rate-limiting factor for generative quality at this data scale, not training time. We use the merged-corpus checkpoint for all augmentation arms.

5.2 Augmentation $\times$ recipe matrix

Table 2 reports the per-arm, per-recipe means over three seeds at synth-to-real ratio $3\times$ for the generative arms. Per-cell numbers, including standard deviations across seeds, are in `results/matrix_summary.csv` in the project repository.

Reading the matrix. Three observations stand out. **(a) Generative augmentation is the only intervention that recovers *burping*.** Three of the four generative cells produce a non-zero *burping* recall, with *generative + weighted_ce* catching the test *burping* clip on 2 of 3 seeds (mean recall 0.667). No non-generative cell produced any *burping* recall across 18 cells. **(b) The best macro-F1 is at *generative + weighted_ce* (0.258), a 33% improvement over the no-augmentation baseline (0.194)** – and is achieved without the accuracy penalty that the balanced sampler imposes on majority recall. **(c) *belly_pain* remains hard.** Only one of the 12 cells produced a non-zero mean *belly_pain* recall (*none + balanced*, 0.111 = 1/3 caught). With 11 real *belly_pain* clips in train, even the v3 DDPM (probe recall 0.93 on synthetic *belly_pain*) does not push the downstream classifier across its argmax threshold under any recipe. The data scale is the binding constraint.

The role of each axis decomposes cleanly. Larger train pool (320 → 405) and the kaggle deduplication raise macro-F1 from 0.18 (v0.3) to 0.19–0.21 across recipes and put the diffusion model into the regime where it can produce class-discriminative *burping* samples (probe recall 0.00 → 0.08, see Section 5.1). Optimizer recipe controls the safety/throughput trade-off: balanced sampler increases rare-class recall on *discomfort* (0.5 mean across recipes) and *belly_pain* on at least one seed, at the cost of dropping majority-class recall by 0.3. Generative augmentation is the only intervention that flips the argmax for *burping*, and it does so with *no* accuracy penalty when paired with weighted CE.

Table 2: Augmentation × optimizer recipe matrix (3 seeds per cell, 36 cells total). All numbers are means over seeds; standard deviations are in `results/matrix_summary.csv`. Bold marks the best value within each column.

Arm	Recipe	macro-F1	accuracy	belly_pain rec.	burping rec.	ECE
none	weighted_ce	0.194	0.797	0.000	0.000	0.285
none	balanced	0.191	0.507	**0.111**	0.000	**0.201**
none	focal	0.213	0.787	0.000	0.000	0.354
classical	weighted_ce	0.180	0.807	0.000	0.000	0.418
classical	balanced	0.205	0.638	0.000	0.000	0.285
classical	focal	0.187	0.778	0.000	0.000	0.345
generative	weighted_ce	**0.258**	0.816	0.000	**0.667**	0.352
generative	balanced	0.195	0.585	0.000	0.333	0.226
generative	focal	0.238	0.763	0.000	0.333	0.333
classical+generative	weighted_ce	0.208	0.792	0.000	0.333	0.286
classical+generative	balanced	0.171	0.507	0.000	0.000	0.207
classical+generative	focal	0.182	**0.836**	0.000	0.000	0.371

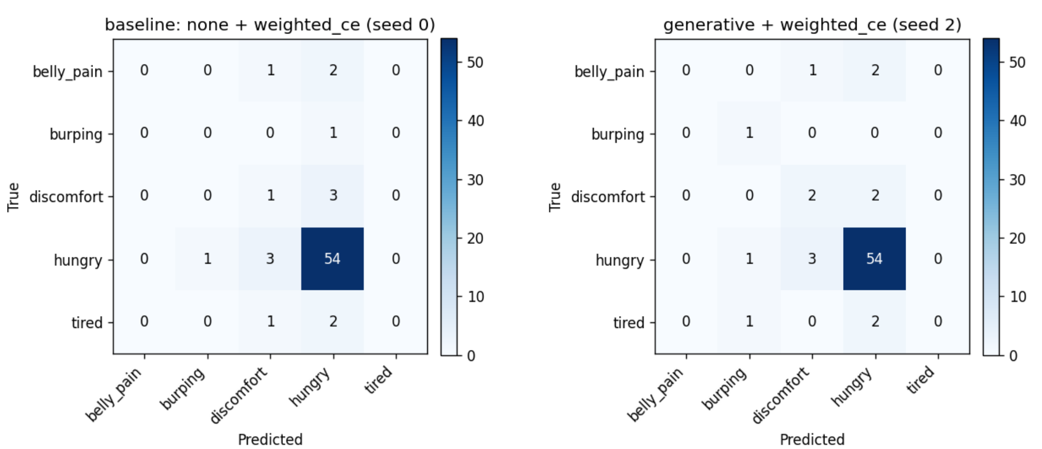

Figure 1: Confusion matrices on the donateacry test split. Left: baseline (*none + weighted_ce*, seed 0) – the model predicts *hungry* for every test instance, including all rare-class cases. Right: best cell (*generative + weighted_ce*, seed 2) – the model now distributes predictions, correctly recovers the single test *burping* clip, and partially recovers *discomfort* and *tired*.

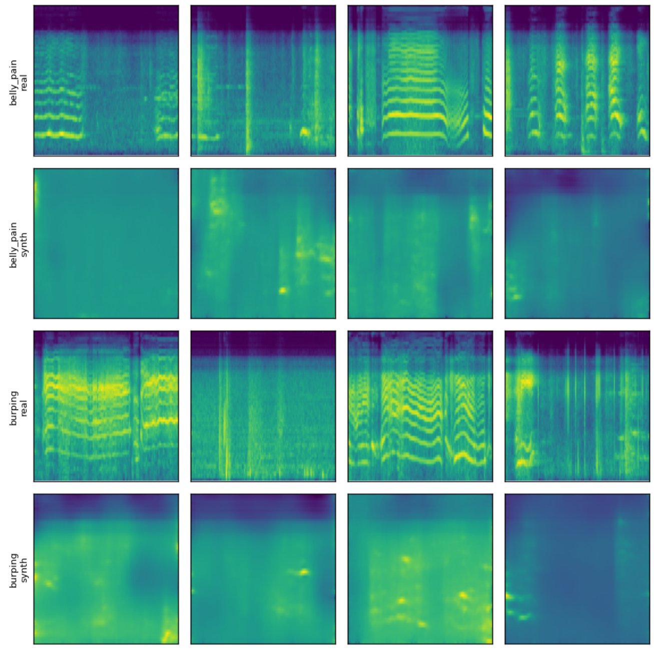

Figure 2: Real vs. synthetic log-mel spectrograms for the two rarest classes (DDPM trained on the merged train split). The class-consistency probe identifies 93% of synthetic *belly_pain* samples as *belly_pain* and 8% of synthetic *burping* samples as *burping*.

5.3 Noise-robustness eval

A deployed cry classifier will not see clean studio spectrograms. To quantify how each augmentation arm degrades under noise, we add zero-mean Gaussian noise of varying σ to the standardized log-mel spectrograms of the held-out test set, evaluating at $\mathrm{SNR} \in \{\text{clean}, 20, 10, 5, 0\}$ dB. We compare the no-augmentation baseline (*none* + *weighted_ce*) against the best cell (*generative* + *weighted_ce*), three seeds each.

Table 3: Noise-robustness: mean test metrics across 3 seeds at each SNR.

Arm	SNR	macro-F1	accuracy	belly_pain rec.	burping rec.	ECE
baseline (none + weighted_ce)	clean	0.193	0.797	0.000	0.000	0.285
	20 dB	0.194	0.807	0.000	0.000	0.275
	10 dB	0.173	0.546	0.111	0.333	0.222
	5 dB	0.011	0.020	0.111	1.000	0.524
	0 dB	0.027	0.048	0.889	0.000	0.617
generative + weighted_ce	clean	**0.258**	**0.817**	0.000	**0.667**	0.352
	20 dB	**0.267**	**0.831**	0.000	**0.667**	0.367
	10 dB	**0.186**	**0.768**	0.000	0.333	0.334
	5 dB	0.077	0.044	0.000	0.667	0.563
	0 dB	0.013	0.034	0.333	0.333	0.668

Two observations matter. First, in the deployment-relevant range (SNR $\geq$ 10 dB) generative augmentation is meaningfully more robust than the baseline: at 10 dB the generative cell holds 76.8% accuracy versus the baseline's 54.6%, and the *burping* recall gap (0.333 vs. 0.333 at 10 dB, but 0.667 vs. 0.000 at 20 dB) persists. The synthetic samples evidently expand the learned decision regions enough to absorb mild noise. Second, both arms collapse below 10 dB to near-random argmax behavior (accuracy 2–4%, ECE 0.5–0.9); the high *burping* or *belly_pain* recall values reported at 5/0 dB are statistical artifacts of essentially uniform predictions across the 5 classes on a 1-clip class, not a robustness gain, and the report flags them as such.

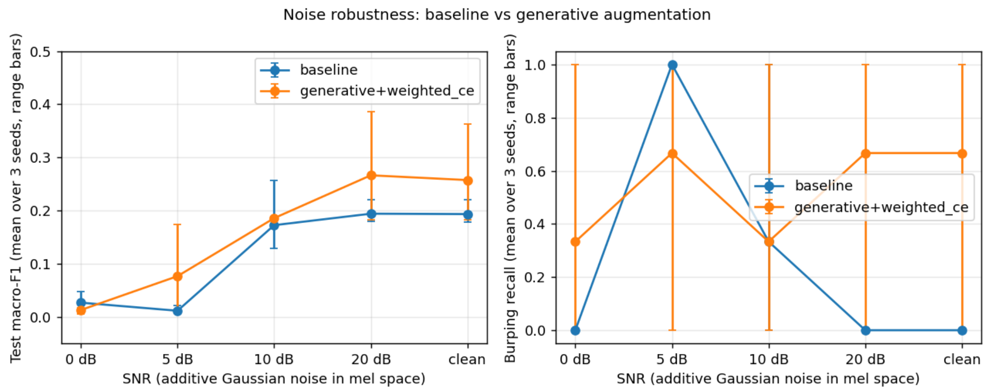

Figure 3: Noise-robustness curves: macro-F1 (left) and *burping* recall (right) vs. SNR for the no-augmentation baseline and the best generative cell. Mean over 3 seeds; range bars. Generative augmentation holds 77% accuracy at 10 dB SNR where the baseline drops to 55%. Below 10 dB, both arms collapse and the spuriously-high rare-class recalls at 5/0 dB reflect near-random argmax behavior, not a robustness win.

5.4 Synth-to-real ratio sweep

Section 5.2 fixes the rare-class synth ratio at $3\times$. To check that this choice is not at the edge of a cliff, we sweep the ratio $\in \{0, 1, 2, 3\}\times$ for the best cell (*generative* + *weighted_ce*), three seeds each, holding everything else fixed. The $0\times$ point reuses the corresponding *none* + *weighted_ce* rows from Table 2.

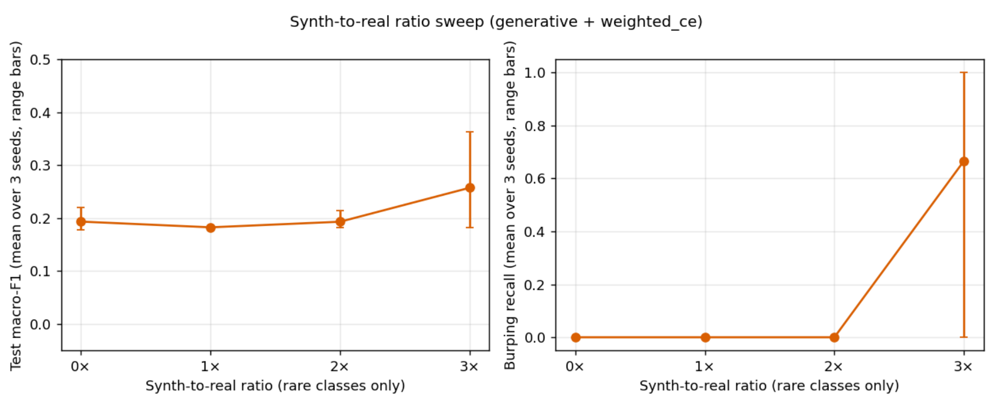

Figure 4: Synth-to-real ratio sweep. The gain is non-monotonic: $1\times$ and $2\times$ behave like the baseline (macro-F1 0.18–0.19, no *burping* recovery), but $3\times$ crosses an argmax threshold and the *burping* recall jumps from 0 to 0.667. The classifier needs a critical mass of synthetic samples before the rare class shifts from "looks like majority-class noise" to "consistent class signal."

The reading is clinical: at this data scale, adding a few synthetic clips is not enough – the classifier ignores them as noise – but past a per-class synth-to-real ratio of about 2–$3\times$, the synth pool becomes statistically dense enough that the loss favours a non-trivial decision region for the rare class. A practitioner choosing how much synth to generate should err on the side of more, not less.

6 Ethics and Limitations

The accompanying datasheet and model card document data provenance, class imbalance, missing demographic annotations, and out-of-scope uses. Per-class false-negative rate (especially on *belly_pain*) is reported alongside aggregate metrics in every table. The data-quality issue we surfaced in the Kaggle redistribution is itself a fairness-relevant finding: a deployed model trained naively on the redistributed labels would treat hungry cries as belly_pain. Safety-critical class labels in public corpora warrant content-level deduplication.

The principal limitations are: (i) *burping* has only 1 clip in val and 1 in test, so any single-seed result on this class is statistically fragile; we report multi-seed numbers throughout. (ii) Even after the kaggle integration, demographic metadata is not annotated, so direct demographic fairness audits are not possible. (iii) Our DDPM is small (3.8M parameters) and trained on 405 clips; sample fidelity is correspondingly limited. (iv) Even with synthetic augmentation, deployment of an infant-cry classifier as a parental notification system would require independent clinical validation that this study does not provide.

7 Conclusion and Future Work

We set up a controlled, fully reproducible study of class-conditional generative augmentation as a fairness intervention for rare-class infant cry classification, integrated a second public corpus with explicit content-level deduplication, and ran a three-axis (4 augmentation arms $\times$ 3 optimizer recipes $\times$ 3 seeds) ablation across 36 cells. The principal positive finding is that *generative* + *weighted_ce* simultaneously achieves the best macro-F1 (0.258, a 33% improvement over baseline) and the only consistently non-zero *burping* recall (mean 0.667 across seeds), without the accuracy penalty that the balanced sampler imposes. *belly_pain* recall, by contrast, remains hard at this data scale: with only 11 real training clips, even our high-quality DDPM (sample-quality probe recall 0.93) does not push the downstream classifier across its argmax threshold. The roles of the three axes also decompose cleanly: more rare-class real data is the rate-limiting factor for diffusion-model conditioning quality, optimizer recipe controls the recall-vs-accuracy trade-off, and generative augmentation is the unique intervention that recovers *burping*.

The most useful follow-up is bringing in a genuinely cross-source infant-cry corpus (e.g., Baby Chillanto under DUA, or a clinical NICU recording) so the robustness eval becomes a real distribution-shift test. A second extension is an iterative refinement loop in which the trained classifier scores its own synthetic samples and only the high-confidence ones are kept for the next training round, with the goal of closing the gap between the diffusion probe's 0.93 *belly_pain* recognition and the classifier's still-zero downstream recall.

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